

DCIT 208 – Software Engineering

D5 :Sprint 1 Review Report and Working Prototype v0.1

Project: ScentHub

Team: InnovateX

Team Repository: <https://github.com/selasi07/ScentHub->

Relevant Commit Hash:

2cc64d968610527cc9c5c79e72f1d28367207d80

DATE: 24th July 2026

Release: v0.1-prototype

A. Sprint Summary

Sprint Goal

Develop the first working prototype of ScentHub that demonstrates a complete product management workflow. The sprint focused on creating a working full-stack application where users can manage perfume inventory using a web interface connected to a database.

Sprint Outcome

Sprint 1 successfully delivered a working prototype with a connected frontend, backend API and SQLite database.

Completed functionality includes:

- User login (prototype authentication)
- View all perfume products
- Add new perfume products
- Edit existing products
- Delete products
- Search products
- SQLite database integration
- Express REST API
- Frontend connected to backend

Sprint Backlog

Story	Status
Login	Completed
View Products	Completed
Add Product	Completed
Edit Product	Completed
Delete Product	Completed
Search Products	Completed
Database Integration	Completed
REST API Development	Completed

No stories were cancelled during Sprint 1.

Sprint Deliverables

Implemented:

- Express backend
- SQLite database
- Product CRUD API
- HTML/CSS/JavaScript frontend
- Product search
- Login prototype
- GitHub repository updates
- Release tag v0.1-prototype

GitHub Evidence

Repository: <https://github.com/selasi07/ScentHub->.

Release Tag: v0.1-prototype

B. Requirements and Backlog Changes

Changes Made

During Sprint 1, the implementation differed slightly from the original architecture proposal.

Database

Originally planned:

- MySQL

Implemented:

- SQLite

Reason:

SQLite allowed faster prototype development without requiring a separate database server while still supporting relational storage.

Frontend

Originally planned:

- React

Implemented:

- HTML
- CSS
- JavaScript

Reason:

A lightweight frontend enabled rapid development within the sprint while demonstrating complete functionality.

Authentication

Originally planned:

- JWT Authentication

Implemented:

- Prototype login using fixed administrator credentials

Reason:

Authentication was deferred to a future sprint to prioritize the inventory management workflow.

No user stories were removed.

Authentication improvements remain in the Product Backlog.

C. Engineering Evidence

Prototype Features

Backend

- Express.js REST API

- SQLite database
- CRUD endpoints

Implemented endpoints:

Method	Endpoint
GET	/products
GET	/products/:id
POST	/products
PUT	/products/:id
DELETE	/products/:id

Frontend

Implemented pages

- Login page
- Inventory dashboard

Implemented features

- Display products
- Add products
- Edit products
- Delete products
- Search products

Testing

Testing completed using Thunder Client.

Verified:

- GET
- POST
- PUT
- DELETE

Manual frontend testing confirmed:

- Product creation
- Product editing
- Product deletion
- Product search
- Database updates

Screenshots

Known Defects

Current prototype limitations

- Login uses prototype credentials instead of database authentication.
- Edit functionality currently uses browser prompt dialogs.
- Product IDs are not reused after deletion due to SQLite AUTOINCREMENT behaviour.
- Customer management has not yet been implemented.
- Sales management has not yet been implemented.

Technical Debt

Remaining work includes

- JWT authentication
- Customer module
- Sales module
- User roles
- Deployment
- Automated testing

- Continuous Integration improvements

D. Metrics and Reflection

Story Points

Planned	Completed
24	24

Sprint completion rate:

100%

Sprint Progress

Sprint 1 burndown

Illustrative sprint burndown based on completed work.

Velocity

Sprint 1 Velocity

24 Story Points

As this is the first sprint, no previous sprint exists for comparison.

Lessons Learned

- Early backend testing simplified frontend integration.
- SQLite accelerated prototype development.
- Frequent Git commits reduced integration problems.
- API testing before frontend development reduced debugging time.
- Incremental implementation made troubleshooting easier.

Contribution Summary

Member	Contribution
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Kelly Adoboe	Backend API, SQLite database integration, frontend implementation, CRUD operations, testing, debugging, GitHub integration and documentation support
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E. Client Feedback

A demonstration of the Sprint 1 prototype was conducted.

Feedback included:

Accepted

- Product management workflow
- Database persistence
- Inventory interface

Suggested Improvements

- Replace prototype login with secure authentication.
- Improve edit interface.
- Add customer management.
- Add sales recording.

These items have been added to the Product Backlog for Sprint 2.

F. Minimum Prototype Requirements

Working User Flow

The following workflow functions end-to-end:

Login

↓

Inventory Dashboard

↓

Add Product

↓

SQLite Database

↓

View Updated Inventory

↓

Edit Product

↓

Delete Product

Local Run Instructions

1. Clone the repository.
2. Install dependencies

npm install

3. Start backend

npm start

4. Open

frontend/index.html

5. Login

Username

admin

Password

admin123

Release Notes

Version

v0.1-prototype

Features

- Working frontend
- Express backend
- SQLite database
- CRUD operations
- Product search
- Prototype login

- GitHub release tag

Conclusion

Sprint 1 successfully delivered the first functional prototype of the ScentHub Perfume Inventory Management System. The project now demonstrates a complete inventory management workflow supported by an Express REST API, SQLite database, and a functional web interface. The prototype provides a solid foundation for Sprint 2, where authentication, customer management, sales recording, automated testing, and deployment will be introduced.